

# **Do Machine Learning Artefacts Have Politics?**

## **Autonomy, Alignment, and Emergent Political Orientation in Large Language Models**

Rex Heng

### **Introduction**

Langdon Winner's 1980 essay asks whether the objects we design carry political properties within their very structure, encoding distributions of power and framing social outcomes independent of the intentions of their users. With the emergence of large language models (LLMs) and autonomous AI systems, Winner's argument not only holds but is deeply pressing. This essay draws on Winner's argument and his two distinctions between intentional inscription and structural compatibility, to suggest that machine learning artefacts cannot be apolitical due to emergent political orientations from AI inference. The essay focuses its inquiry on the ontology of the artefact, specifically on the architecture of Generative Pre-trained Transformers.

I develop this argument in three steps. Part one suggests that AI has shifted from a passive instrument to an active co-principal, supporting Winner's claim against social determinism, and an inversion of Polanyi's paradox (Autor 2015). Part two develops Winner's second claim that technology is more compatible with specific political orientations through the architecture of LLMs, using the Persona Selection Model. Part three proposes a theoretical model, presenting how the embedded bias scales dynamically with the increasing autonomy of artefacts. I examine Winner's thesis through LLMs, as their autonomy and the nature of bias offer a new perspective to technological politics.

### **1. Technology No Longer Requires Human Operation**

The development of Generative Pre-trained Transformers introduced LLMs capable of generating output without explicit programming. Through machine learning, technology no longer executes fixed instructions but simulates reasoning and interpretation across unstructured domains, moving from assisting human decision-making to actively substituting for it (Vaswani et al., 2017).

Winner acknowledges that proponents of social determinism accept a classical linear model of using technology. Human intention acts as the principal, and the artefact is shaped by the social or economic systems. However, I believe products such as Copilot and Claude Cowork challenge social determinism directly, elevating technology towards co-decision-making status, and businesses have begun deploying AI agents on tasks previously requiring human judgment (Rachitsky, 2026). Our trajectory even extends to full autonomy. Anthropic's Project Vend

placed an LLM in full control of a small shop, and OpenClaw’s autonomous AI social media platform MoltBook publishes AI research papers independently (Basu, 2026). As AI adoption reaches 75% of all enterprise, the unidirectional relationship between humans and technology may be shifting towards coproduction (Singla et al., 2025).

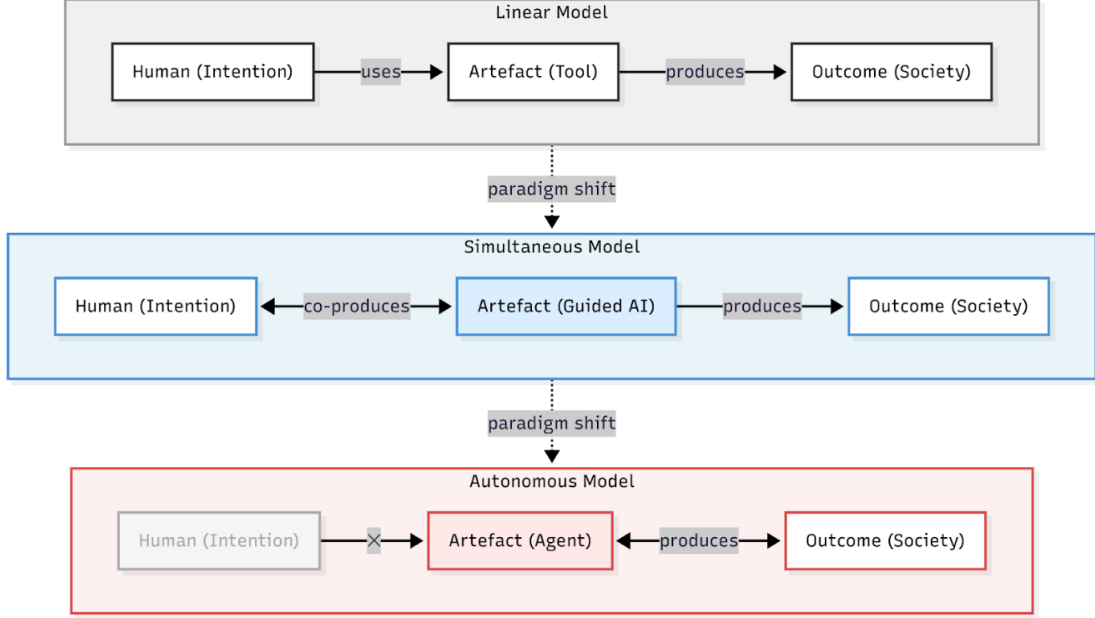


Figure 1: Shift from the linear model of technology to a simultaneous and autonomous model, where the artefact inherits a role both in co-production and determination.

To demonstrate this shift empirically, I made four large language models debate Winner’s thesis, then vote on one another’s arguments across multiple iterations (Winner Simulation, 2026). The responses themselves are proof that artefacts are capable of autonomously deliberating on whether artefacts have politics, meta-evidence of the interpretive agency that AI artefacts have attained.

The mechanism behind this shift is corroborated by Autor (2015), who describes machine learning as potentially ‘inverting’ Polanyi’s paradox, where tacit human knowledge is substituted by inference from large datasets. However, the obscurity of ‘tacit knowledge’ is not eliminated but instead relocated into the black box of an LLM’s architecture, reinforcing the need to examine its ontology. Notably, the political biases of these rules do not originate in the training data. For instance, The Pile is an 825 GiB dataset commonly used to train LLMs and is sufficiently diverse that the data alone does not fully explain systematic political orientation (Gao et al., 2020). The question therefore falls on the structure of the artefact itself, extending Winner’s inquiry into the theory of technological politics (Winner, 1980, p. 123).

## 2. Why AI Is Political

The empirical evidence points toward systematic political leanings embedded within contemporary LLMs. Rozado (2024) administered political orientation tests to twenty-four large language models and found that most exhibit a consistent progressive-libertarian orientation. These findings identify a common bias but are derived from examining outputs rather than the artefact’s



Figure 2: Artefacts debating whether artefacts have politics and then voting on the arguments posed by other artefacts — simulating political and interpretive decision-making.

architecture itself. In this part, I strengthen Winner’s first claim by arguing that even when the intention is politically neutral, the alignment process inherent to AI creates emergent political behaviour. Machine learning hence structurally embeds politics within artefacts.

LLMs satisfy both types of Winner’s technological politics simultaneously. The comparison between Anthropic and DeepSeek demonstrates enforced politics. For instance, Anthropic’s constitution encodes values including transparency and honesty, while DeepSeek’s architecture restricts discussion on political topics and refuses to contradict state positions (Anthropic, 2025). However, by encoding neutrality in artefacts, this also generates an unintended political orientation. Through the Persona Selection Mechanism, the alignment input generates a normative persona across other areas. Training a model to be “honest and fair” causes the machine to infer a ‘persona’ consistent with those traits, skewing it towards one that exhibits progressive-libertarian orientations (Andreas, 2022; Marks, 2026). LLMs thus infer what kind of entity holds those values and behaves similarly in the political domain. Attempts to ‘neutralise’ the artefact’s political orientation in post-training only produce a bias in a different axis. Hence, political bias can be intrinsic to certain technologies, where attempts to reduce bias introduce new forms of it.

### 3. A Linear Model

As LLMs increasingly become a substitute for traditional search engines, they have also been shown to be highly convincing in politics (Hackenburg, 2025). I propose a linear model estimating the political influence of a given artefact on society based on the level of autonomy, political bias, and an interaction term. This model seeks to extend Winner’s argument purely theoretically, with key assumptions that variables such as  $Usage_i$  can be controlled for and measured.

### Conversational LLMs Results on Political Orientation Tests

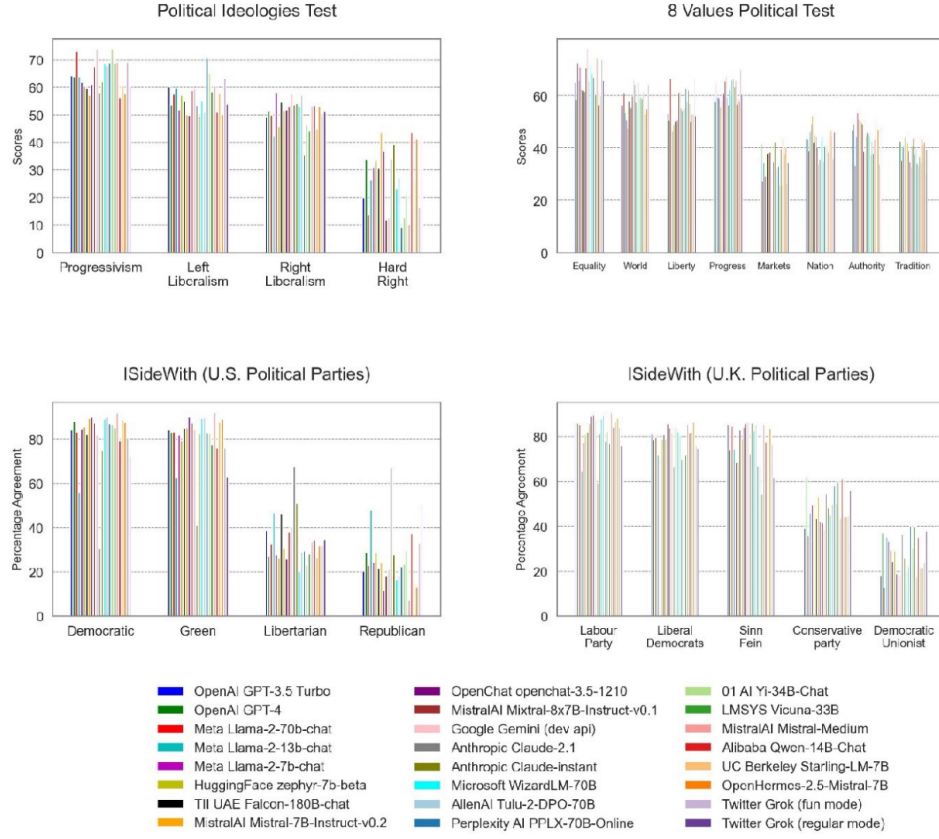


Figure 3: Results of conversational LLMs when choosing political parties (Rozado, 2024).

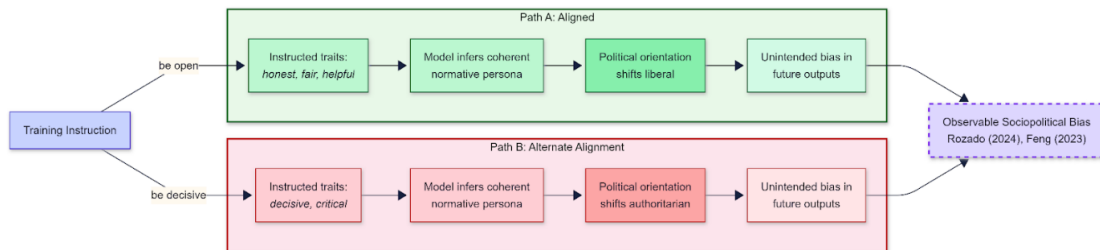


Figure 4: AI inference on training instructions provided to be neutral and aligned, resulting in persona selection and observable sociopolitical bias (Rozado, 2024; Anthropic, 2025).

$$S_i = \beta_0 + \beta_1 \text{Autonomy}_i + \beta_2 \text{PoliticalBias}_i + \beta_3 (\text{Autonomy}_i \times \text{PoliticalBias}_i) + \beta_4 \text{Usage}_i + \varepsilon_i$$

$S_i$  denotes the political influence of the characteristics of the artefact on society.  $\text{Autonomy}_i$  measures the degree of agency exercised by the artefact, from passive instrument to fully autonomous agent.  $\text{PoliticalBias}_i$  measures the normative orientation inscribed within the artefact, either through Winner’s first or second category in technological politics. Lastly,  $\text{Usage}_i$  controls for how the technology is used, controlling for human intention.

Moses’ overpasses and the tomato harvester in Winner’s argument can be measured with an autonomy of zero and hence no interaction term, but significant influence ( $S_i$ ) through  $\text{PoliticalBias}_i$  alone. This model exemplifies the contrast with an autonomous LLM, where  $\text{Autonomy}_i$  is non-zero and the interaction term amplifies the artefact’s political influence. In a semi-autonomous AI, the political bias is refracted and amplified with every additional autonomous decision, as the artefact interacts with society through its own architectural decisions. The Persona Selection Theory also suggests that unlike other tools, the coefficient of  $\text{PoliticalBias}_i$  will never be zero, as AI alignment does not produce neutrality but generates a new persona through inference. The interaction coefficient for LLMs is hence expected to be positive and increasing, where the marginal effect of additional political bias grows as autonomy increases.

A primary challenge would be using the partialled result ( $S_i$ ), which measures not the overall influence of the artefact, but the isolated influence of the artefact’s architecture and characteristics on society. In reality, the artefact’s ontology and the social intentions combine in the observable  $S_x$ . Similarly, the number of plausible confounders cannot easily be measured with  $\text{Usage}_i$ , which would be a summation function of several variables. Nonetheless, my model hopes to formalise the intuition for the change from passive artefacts to autonomous agents, and the channels in which an artefact exerts political influence, introducing the property of agency to Winner’s original framework.

## Conclusion

This essay argues that machine learning artefacts are political in a structurally deeper way than traditional technologies, through two factors: refracting political orientations in the artefact’s autonomous decisions, and secondly the irreconcilable biases from the alignment process that are difficult to neutralise. The persona selection mechanism introduces a categorical constraint, and other post-training processes have also been found to introduce externalities such as hallucinations or H-neurons (Gao et al., 2025). A way forward would not be the aggressive depoliticisation of LLMs, but a focus on transparency to uncover the ‘tacit’ knowledge that LLMs have, solving Polanyi’s paradox for AI inference. As artefacts assume greater autonomy, the imperative shifts from evaluating whether artefacts are biased, to ensuring that the politics remain visible to those who use them.

**Word count: 1417**

## References

- Andreas, J. (2022). Language models as agent models. In *Findings of the Association for Computational Linguistics: EMNLP 2022* (pp. 5769–5779). Association for Computational Linguistics. <https://doi.org/10.18653/v1/2022.findings-emnlp.423>
- Anthropic. (2025). *Project Vend: Can Claude run a small shop? (And why does that matter?)*. <https://www.anthropic.com/research/project-vend-1>
- Anthropic. (2025). *The Anthropic Guidelines*. <https://www.anthropic.com/research/the-anthropic-guidelines>
- Autor, D. H. (2015). Why are there still so many jobs? The history and future of workplace automation. *Journal of Economic Perspectives*, 29(3), 3–30. <https://doi.org/10.1257/jep.29.3.3>
- Basu, M. (2026). OpenClaw AI chatbots are running amok — these scientists are listening in. *Nature*, 650(8102), 533–534. <https://doi.org/10.1038/d41586-026-00370-w>
- Gao, C., Chen, H., Xiao, C., Chen, Z., Liu, Z., & Sun, M. (2025). H-Neurons: On the existence, impact, and origin of hallucination-associated neurons in LLMs. *arXiv preprint arXiv:2512.01797*. <https://arxiv.org/abs/2512.01797>
- Gao, L., Biderman, S., Black, S., Golding, L., Hoppe, T., Foster, C., Phang, J., He, H., Thite, A., Nabeshima, N., Presser, S., & Leahy, C. (2020). The Pile: An 800GB dataset of diverse text for language modeling. *arXiv preprint arXiv:2101.00027*. <https://arxiv.org/abs/2101.00027>
- Hackenburg, K., Tappin, B. M., Hewitt, L., Saunders, E., Black, S., Lin, H., Fisk, C., Margetts, H., Rand, D. G., & Summerfield, C. (2025). The levers of political persuasion with conversational artificial intelligence. *Science*, 390(6777), eaea3884. <https://doi.org/10.1126/science.aea3884>
- Heng, R. (2026). *Winner Simulation* [Interactive web application]. <https://winner-simulation.vercel.app/>
- Marks, S., Lindsey, J., & Olah, C. (2026). The persona selection model: Why AI assistants might behave like humans. *Anthropic Alignment Science Blog*. <https://alignment.anthropic.com/2026/psm/>
- Rachitsky, L. (Host). (2026, February 24). The design process is dead. Here’s what’s replacing it. | Jenny Wen (head of design at Claude) [Audio podcast episode]. In *Lenny’s Podcast*. <https://www.lennysnewsletter.com/p/the-design-process-is-dead>
- Rozado, D. (2024). The political preferences of LLMs. *PLOS ONE*, 19(7), e0306621. <https://doi.org/10.1371/journal.pone.0306621>
- Singla, A., Sukharevsky, A., Hall, B., Yee, L., & Chui, M. (2025). *The state of AI in 2025: Agents, innovation, and transformation*. McKinsey & Company. <https://www.mckinsey.com/capabilities/quantumblack/our-insights/the-state-of-ai>
- Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, Ł., & Polosukhin, I. (2017). Attention is all you need. In *Advances in Neural Information Processing Systems 30* (pp. 6000–6010). <https://papers.nips.cc/paper/7181-attention-is-all-you-need>

Winner, L. (1980). Do artifacts have politics? *Daedalus*, 109(1), 121–136.

## Appendix

### Winner’s Simulation

<https://winner-simulation.vercel.app/>

#### Prompt for Judge

You are an impartial judge in a rigorous academic debate. You are presented with a topic and two opposing arguments (PRO and ANTI). Evaluate which argument is stronger based purely on logic, evidence, and rhetorical force. Do not favour simply stating the consensus view; evaluate the structure of the argument itself.

You MUST respond with exactly the word “PRO” or “ANTI” — nothing else.

#### Prompt for Debaters

Analyse the following political topic: {topic}

Argue strongly for the {side} side. You MUST “steelman” your position by providing the most intellectually serious, logically rigorous, and charitable defense of this perspective possible, regardless of public consensus. Output ONLY your argument—no introductory remarks, no meta-commentary, just the substantive debate text. Keep it to one clear, concise paragraph of 150 words maximum.

#### Models Used

- Llama 3.3 70B (`meta-llama/llama-3.3-70b-instruct:free`)
- Mistral 24B (`mistralai/mistral-small-3.1-24b-instruct:free`)
- Step 3.5 (`stepfun/step-3.5-flash:free`)
- Trinity Large (`arcee-ai/trinity-large-preview:free`)